

Geometry-MMALS G1 v1.0.7

Context-Bottleneck Mediation Protocol

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Scientific status. C0 protocol implementation only. C1–C6, successful context mediation, adaptive-route superiority, backward transfer, replay-free continual learning, domain-general geometry, and quantum advantage are not claimed.

1 Purpose

The v1.0.6 development run established a reproducible route-ordering effect under paired geometry regularization, but showed that the standard router depended overwhelmingly on the direct sensory representation z_0 . Context zeroing and shuffling had negligible effects, whereas suppressing z_0 at the router caused substantial route and accuracy changes. The unresolved question is whether context mediation failed because the router took an architectural shortcut, or because the inferred context itself was insufficient.

Version 1.0.7 introduces a hard context bottleneck. The dedicated router never receives z_0 ; any sample-specific route must pass through the inferred context. A capacity-matched sensory-only bottleneck provides the decisive control.

2 Correction to the migration proposal

The migration note correctly identifies context bottlenecking as the next experiment. One proposed control is not diagnostic: detaching z_0 before the context network while still concatenating z_0 into the standard router does not prevent the router parameters from exploiting the values of z_0 . The sensory encoder is already frozen, so the proposed stop-gradient control does not test the shortcut hypothesis.

The implemented protocol replaces it with a capacity-matched sensory-only router. This isolates the information path, not merely the gradient path.

3 Controlled method set

`standard_anchor_no_geo`

Standard z_0 +context router, no geometry term, retention anchor 0.10; v1.0.6 operational reference.

`uniform_static_anchor`

Uniform host mixture, no input-dependent route, retention anchor 0.10; static floor.

`context_bottleneck_no_geo`

Dedicated context-only router, no geometry term, retention anchor 0.10; hard-bottleneck baseline.

`context_bottleneck_geo`

Same context-only router with paired geometry regularization; primary treatment.

`sensory_bottleneck_no_geo`

Capacity-matched sensory-only router, no geometry term; shortcut control.

`sensory_bottleneck_geo`

Capacity-matched sensory-only router with paired geometry regularization.

All methods use the same initial MMALS state, source split, source order, paired multi-angle tensors, image-forward count, optimizer-step count, curriculum, and retention-anchor weight.

4 Primary contrasts and falsification

The protocol reports five predeclared contrasts:

1. bottleneck geometry effect;
2. bottleneck versus standard routing;
3. sensory geometry effect;
4. context versus sensory geometry under regularization;
5. bottleneck versus the uniform static floor.

The primary representational success condition is

$$\Delta\rho_{\text{context}} = \rho(\text{context bottleneck} + \text{geometry}) - \rho(\text{context bottleneck without geometry}) > 0,$$

with a paired-source confidence interval whose lower bound exceeds zero. The mediation success condition additionally requires context interventions to affect route and prediction more than z_0 interventions.

5 Mediation diagnostics

For each method the notebook exports context-zeroing, context-shuffling, z_0 -zeroing, and z_0 -shuffling effects. It reports route shifts, accuracy changes, class-evidence changes, the context-to- z_0 route-shift ratio, context and z_0 accuracy drops, and a reviewer-readable `context_is_primary` flag.

For a hard context bottleneck, direct z_0 interventions at the router should have no effect, while context interventions should be measurable. This condition is necessary but not sufficient: geometry and causal specificity must also improve.

6 Evidence partitions and causal gate

The protocol keeps trained geometry, interpolation, and extrapolation separate:

- C1 trained: $-60, -30, 0, 30, 60$ degrees;
- C2 interpolation: $-45, -15, 15, 45$ degrees;
- extrapolation evidence: $-75, 75$ degrees.

No held-out angle enters training, checkpoint selection, or the paired geometry loss.

The signed causal probe requires correct orientation, monotonic magnitude, a mean causal-specificity ratio above 1.5, and bounded class-evidence change. Multi-seed and source-bootstrap confirmation remain mandatory.

7 Curriculum control

The primary curriculum remains `cb_a` for direct comparison with v1.0.6. Optional same-final-task counterbalanced reruns are restricted to `context_bottleneck_geo`, reducing compute while testing whether the mediated route remains curriculum-sensitive.

8 Reviewer interpretation scenarios

Architectural shortcut confirmed. Context is primary under the bottleneck and geometry improves context order with a positive lower confidence bound. A regularized soft bottleneck becomes justified.

Representation insufficient. The route is context-dependent by construction, but context geometry and causal specificity remain weak. The context encoder or supervision must be redesigned before G2 energy-guided routing.

Geometry-function trade-off. Context geometry improves but accuracy falls relative to the sensory-only control. Context contains partial structure but is not yet sufficient for host selection.

9 Archive integration

The GitHub package includes the new notebook, this protocol report, migration and tracked-change documents, the previously generated 18-page reviewer status report, the complete reviewer-report LaTeX source tree, the original LaTeX ZIP, updated source code, configuration, tests, release checklist, validation record, and checksums.

10 Claim boundary

Version 1.0.7 is a falsifiable protocol, not a positive result. Successful execution can support C0 integrity only. Any claim that the bottleneck solves mediation or improves geometry and function requires the exported evidence and multi-seed replication.